Explaining the Performance Plateau in Machine Learning-based Juvenile Recidivism Prediction on PACT dataset

Anish Sandaka
University Of North Texas
Denton, Texas, USA
AnishSandaka@my.unt.edu

Jessica Craig
University Of North Texas
Denton, Texas, USA
Jessica.Craig@unt.edu

Yunhe Feng
University Of North Texas
Denton, Texas, USA
yunhe.feng@unt.edu

Chad Trulson
University Of North Texas
Denton, USA
Chad.Trulson@unt.edu

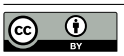
Abstract

Predicting whether a juvenile will reoffend is an important problem in juvenile justice. The Positive Achievement Change Tool (PACT) is commonly used to assess juvenile risk and needs, but machine learning models based on PACT data often achieve Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) values of only about 0.61–0.72. This paper investigates whether this performance plateau is caused by the machine learning pipeline or by limitations in the information available in the PACT assessment. We compared several machine learning algorithms and encoding strategies, expanded the feature set using Risk-Need-Responsivity (RNR) theory, and added county-level socioeconomic and community variables. We also used SHapley Additive exPlanations (SHAP) to examine which features contributed most to prediction. The results showed that changing algorithms, encoding strategies, and PACT feature sets produced only small changes in ROC-AUC. Adding county-level information resulted in a small but consistent improvement across repeated evaluations. SHAP analysis showed that attitudes/skills and criminal history were among the most important feature groups, while the overall PACT intake risk score contributed little once the individual assessment items were included. These results suggest that the prediction ceiling is mainly related to limitations in the information available at intake rather than the choice of machine learning method.

Keywords

Juvenile recidivism, Risk assessment, PACT, Explainable AI, Risk-Need-Responsivity (RNR)

ACM Reference Format:
Anish Sandaka, Jessica Craig, Yunhe Feng, and Chad Trulson. 2026. Explaining the Performance Plateau in Machine Learning-based Juvenile Recidivism Prediction on PACT dataset. In *The 2026 ACM/IEEE Joint Conference on Digital Libraries (JCDL '26), October 13–16, 2026, Frisco, TX, USA.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3805696.3847299



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JCDL '26, Frisco, TX, USA
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ACM ISBN 979-8-4007-2597-5/2026/10
https://doi.org/10.1145/3805696.3847299

1 Introduction

Juvenile justice agencies use structured risk assessments to inform decisions about placement, supervision, and treatment. One instrument used in Texas is the Positive Achievement Change Tool (PACT) [8, 9], which was adapted from the Washington State Juvenile Court Assessment [5, 13] and subsequently validated in Florida [3, 13]. Because these assessments inform consequential decisions, machine learning models developed from PACT data require rigorous and consistent out-of-sample evaluation.

Previous studies using PACT and related juvenile risk assessments have generally reported ROC-AUC values between 0.61 and 0.72 [2, 3, 8, 9, 13]. The persistence of this performance range raises an important question: does the apparent plateau reflect limitations in the machine learning pipeline, or does it reflect limitations in the information captured by the assessment itself? Existing studies often evaluate a specific model and feature set, providing comparatively limited evidence about why performance remains within this range. A controlled comparison of modeling choices is therefore needed to determine whether changes in algorithms, feature encoding, or available predictors can meaningfully improve predictive performance.

To address this question, we systematically varied the learning algorithm, feature-encoding strategy, and feature set. Except for the exploratory baseline in Table 1, all controlled comparisons used the same cross-validation framework, with preprocessing fitted separately within each training fold before evaluation on the corresponding validation fold. This ensured that all configurations were compared under equivalent conditions. We organized the feature sets according to Risk-Need-Responsivity (RNR) theory [1] and consultations with a juvenile justice domain expert. Model A included a core set of theoretically selected variables; Model B extended this set with additional criminogenic factors; and Model C further incorporated non-criminogenic factors. We also evaluated a broader feature expansion to test whether adding a larger number of intake variables, without regard to their theoretical classification, improved performance. These experiments used PACT data provided by the Texas Juvenile Justice Department (TJJD).

After evaluating the information available within PACT, we examined whether contextual information from outside the assessment offered additional predictive value. County-level socioeconomic, demographic, and community-health variables from the U.S. Census Bureau's American Community Survey (ACS) and County

Health Rankings (CHR) were linked to each juvenile using the headquarters county of the corresponding probation department. Because the resulting performance gains were modest, we assessed their consistency through a paired 10-seed evaluation: for each seed, the baseline and county-augmented models were trained and evaluated using identical cross-validation splits. Finally, we used SHapley Additive exPlanations (SHAP) to identify the feature domains that contributed most strongly to model predictions and to assess the contribution of county-level context beyond the information contained in PACT.

Across experiments restricted to PACT variables, changes in algorithms, encoding strategies, and feature sets produced only minor differences in ROC-AUC. Adding county-level contextual variables, however, yielded a small but consistent improvement across repeated evaluations. The SHAP analysis identified attitudes and skills, together with criminal history, as influential predictor domains and indicated that county-level context contributed information not captured by the PACT variables. Taken together, these findings suggest that the observed performance ceiling is more closely associated with limitations in the information available at intake than with the choice of machine learning method.

This work makes three primary contributions. First, it uses a controlled cross-validation design to compare algorithm selection, feature encoding, and feature-set composition under consistent evaluation conditions. Second, it applies RNR theory to compare criminogenic and non-criminogenic feature expansions and evaluates whether external community context provides incremental predictive value. Third, it combines paired, multi-seed evaluation with SHAP-based interpretation to assess both the consistency of the improvement from external context and the types of information driving model predictions.

2 Related Work

Statistical and machine learning approaches have been widely applied to juvenile recidivism prediction. Studies using PACT and comparable structured risk assessments have generally reported ROC-AUC values between 0.61 and 0.72 [2, 3, 8, 9, 13]. PACT was adapted from the Washington State Juvenile Court Assessment [5] and was subsequently validated using data from Florida [3, 13]. Related work has also examined PACT-based risk assessment in Texas juvenile justice settings [8, 9]. Whereas these studies primarily evaluate the predictive validity of PACT, our paper investigates why predictive performance remains within a relatively narrow range. In particular, we examine whether this apparent performance ceiling can be attributed to choices within the machine learning pipeline or to limitations in the information available at intake.

Our feature-set design is informed by the Risk-Need-Responsivity (RNR) framework, which distinguishes criminogenic needs—dynamic factors directly associated with offending—from non-criminogenic factors that may be relevant to treatment or case management but are generally less predictive of recidivism [1]. Using this distinction, we organize PACT variables into theoretically motivated feature groups and evaluate the incremental predictive value of adding criminogenic and non-criminogenic factors. This approach allows us to assess not only whether additional variables improve performance, but also whether theoretically relevant feature expansions are more informative than broader, less selective expansions.

Prior research further suggests that socioeconomic conditions and neighborhood disadvantage may provide contextual information beyond individual-level risk factors [4]. To evaluate the contribution of such information, we augment the PACT variables with county-level measures from the U.S. Census Bureau's American Community Survey (ACS) and County Health Rankings (CHR). These data enable us to test whether external community context explains variation in recidivism risk that is not captured by the individual assessment.

Taken together, the existing literature establishes the predictive validity of PACT and related instruments but provides less evidence about the source of their apparent performance limitations. Our study addresses this gap by comparing learning algorithms, encoding strategies, theoretically organized feature sets, and external contextual variables within a common cross-validation framework. This design helps distinguish limitations arising from model specification from those arising from the information available to the models.

3 Methodology

3.1 Overview

We designed this study as a controlled comparison to determine whether the performance plateau in PACT-based juvenile recidivism prediction arises primarily from choices in the machine learning pipeline or from limitations in the information available at intake. Within a common cross-validation framework, we varied three components of the prediction pipeline: the learning algorithm, the feature-encoding strategy, and the feature set. The feature-set experiments progressively incorporated criminogenic and non-criminogenic variables based on Risk-Need-Responsivity (RNR) theory.

We first compared configurations restricted to PACT variables. We then augmented Model B with county-level socioeconomic, demographic, and community-health variables linked through the headquarters county of the supervising juvenile probation department. Because the observed improvement from these contextual variables was modest, we evaluated its consistency using a paired, multi-seed protocol in which the baseline and county-augmented models were evaluated on identical data splits.

3.2 TJJD PACT data

We used the TJJD PACT dataset `TJJDPACT040226.dta`, which initially contained 4,646 assessment records and 748 variables. Structural cleaning was performed once before model evaluation, whereas all preprocessing operations that estimated parameters from the data, including imputation, scaling, and encoding, were fitted separately within each training fold.

During the initial data audit, we removed variables with severe missingness or near-constant values. We also removed three records in which `D1Q1`, representing age at first offense, was coded as "None," which was identified as a data-entry artifact. Duplicate `PIDNumber` entries were resolved by retaining the most recent assessment for each individual, removing 82 redundant records. Blank strings were

converted to missing values but were not imputed at this stage. The final analytic dataset contained 4,561 individuals, with one assessment record per individual.

The binary outcome, `Recidivate_max`, indicates whether an individual had any recorded reoffense without a fixed follow-up horizon. The outcome distribution was 74.6% positive and 25.4% negative. For age, we used `ageatincar` based on guidance from the juvenile justice domain expert. We excluded `Agerelease` because it is partly determined by sentence length and therefore would not constitute an appropriate intake-time predictor.

3.3 Feature groupings: Models A, B, and C

We specified three nested feature configurations a priori using RNR theory [1] and consultation with a juvenile justice domain expert. Variables were not added or removed based on their predictive performance in the current dataset. This design allowed us to evaluate the incremental contribution of theoretically defined feature domains while holding the remaining components of the modeling pipeline constant.

Model A (44 variables) contains the core theory-selected PACT variables, including measures of criminal history, attitudes, peer relationships, parental control, intake risk, demographics, and selected ordinal and continuous characteristics. The initial specification listed 45 variables; however, a subsequent audit showed that `Impulsivity` was an exact recoding of `D10Q4`. We therefore removed `Impulsivity`, leaving 44 unique variables. Removing this duplicate did not affect model performance.

Model B (87 variables) extends Model A by adding criminogenic variables from domains such as family functioning, school experiences, cognitive skills, use of free time, prior convictions, and consequences of substance use.

Model C (130 variables) extends Model B by adding non-criminogenic variables related to mental health, abuse and neglect, and family social history.

Because the configurations are nested, comparisons among Models A, B, and C measure whether progressively broader theoretically defined information provides predictive value beyond the preceding feature set.

3.4 Theory-driven feature encoding

To avoid using the outcome to define predictor representations, we specified ordinal category orderings using criminological theory rather than recidivism rates observed in the current dataset. Under the theory-driven encoding strategy, 28 categorical variables were ordinally encoded and 9 were one-hot encoded. Continuous and ordinal-numeric variables were median-imputed and scaled within each training fold.

For ordinal variables, category levels were ordered from lower to higher theoretical risk based on criminological knowledge. Empirical recidivism rates were inspected only as a diagnostic after the ordering had been specified; they were not used to construct or revise the category order. This approach reduces the risk of producing an encoding that is overly specific to the current sample.

To evaluate the sensitivity of model performance to the encoding strategy, we considered three encoding approaches: the theory-driven hybrid encoding, a simpler uniform ordinal encoding, and the one-hot encoding used in the exploratory baseline. The controlled encoding comparison was between the theory-driven and uniform ordinal approaches, with the remaining evaluation conditions held constant. For the Random Forest model, the absolute difference between these two approaches was approximately 0.015 ROC-AUC. The one-hot results in Table 1 were obtained using a different feature set and an exploratory single 75/25 split and therefore are not treated as a direct encoding comparison. These results indicate that the evaluated encoding choice had only a limited effect on performance. The encoding strategy was held constant when directly comparing Models A, B, and C so that differences among these configurations could be attributed to feature content rather than to changes in encoding.

3.5 Leakage control

We defined the prediction point as the time of intake and excluded variables that would not reasonably be available at or before that point. In total, we removed 169 variables. Of these, 168 were timing-related exclusions: 141 institutional-misconduct variables with the `_sum_max` designation, which describe in-custody behavior; 5 sentence-related variables, namely `Agerelease`, `sentencetype`, `postLDA_max`, `deterprisontransfer`, and `PRISONTRANSFERORCERT_max`; and 22 variables related to post-release outcomes or time in custody, including `postarrests_sum`, `TYCtime_max`, and `threeyearfelonyrecid`. The remaining variable, `preLDAscale_max`, was excluded because its Stata metadata label indicated that it was derived from recidivism outcome information. These exclusions ensured that the models used only information plausibly available at the intended prediction time. Including post-intake or outcome-derived information could otherwise inflate ROC-AUC and produce an unrealistic estimate of prospective model performance.

3.6 Modeling pipeline and evaluation

All models were implemented using scikit-learn pipelines [10]. Within each cross-validation fold, imputation, encoding, and scaling were fitted exclusively on the training partition and then applied to the held-out partition. Consequently, no information from the held-out observations was used to estimate preprocessing parameters.

In this study, binary 0/1 variables were imputed but not scaled. Continuous and ordinal-numeric variables were median-imputed and scaled within each training fold, and categorical variables were processed using the corresponding encoding strategy.

We evaluated predictive performance using stratified 5-fold cross-validation, with ROC-AUC as the primary metric. For each configuration, we report the mean ROC-AUC and, where available, the standard deviation across the five folds. When comparing algorithms, encoding strategies, or feature configurations, we used the same fold assignments and random seed. This paired design ensured that observed performance differences were not attributable to differences in how the data were partitioned.

3.7 External context and multi-seed evaluation

To test whether information not captured by PACT could improve prediction, we augmented Model B with county-level socioeconomic, demographic, and community-health variables from the U.S.

Census Bureau's American Community Survey and County Health Rankings. These variables were linked using `HQCountyNumber`, which identifies the headquarters county of the supervising juvenile probation department. All 148 county codes represented in the TJJD dataset were successfully mapped to Texas counties.

The headquarters county may differ from an individual's county of residence. The linked variables should therefore be interpreted as measures of the administrative and community context associated with the supervising probation department rather than as direct measurements of each individual's residential neighborhood. This geographic approximation is a limitation of the analysis.

To determine whether the modest improvement associated with the county-level variables was consistent across data partitions, we repeated stratified 5-fold cross-validation using 10 random seeds. For each seed, Model B and the county-augmented Model B+Geo were evaluated using identical fold assignments and otherwise identical modeling procedures. The two configurations therefore differed only in the inclusion of the county-level variables.

For each seed s, we first calculated the mean ROC-AUC across its five folds and then computed the paired improvement as $\Delta_s = \mathrm{AUC}_{s,\mathrm{B+Geo}} - \mathrm{AUC}_{s,\mathrm{B}}$. We summarize the paired evaluation using the mean and standard deviation of Δ_s across the 10 seeds, together with the number of seeds for which $\Delta_s > 0$. This procedure evaluates whether the contribution of county-level context is consistently positive rather than being driven by a single favorable partition of the data.

4 Experiments and Results

4.1 Experimental setup

All experiments were implemented in Python 3.11 using scikit-learn 1.5 [10]. Random Forest was trained with 500 trees. HistGradientBoosting used the default learning rate and maximum number of iterations, while Logistic Regression used L2 regularization with $C = 1.0$ and `max_iter` = 1000. For the internal experiments, the random seed was fixed at 42. For the external county-level experiments, we repeated the evaluation using random seeds $\{0, 1, \ldots, 9\}$. All preprocessing was performed within a `ColumnTransformer`, ensuring that imputation, encoding, and scaling were fitted only on the training data in each fold.

4.2 Baselines

We begin by comparing six machine learning models to determine whether algorithm choice has a meaningful impact on prediction performance. Table 1 shows the results on the initial 48-feature set using one-hot encoding.

The results show: the best and worst tree-based models are separated by only 0.08 in ROC-AUC, and the top three algorithms—spanning bagged trees, boosted trees, and linear models—differ by at most 0.011 in ROC-AUC. The KNN baseline's high accuracy is an artifact: it predicts "reoffend" for ∼97% of cases, and its accuracy simply reproduces the base rate. From this point forward we drop KNN, Naive Bayes, and Decision Tree as uncompetitive and standardize on Logistic Regression, Random Forest, and HistGradientBoosting — three families with different inductive biases, ensuring that our downstream comparisons are not artifacts of one algorithm.

Table 1: Baselines on the initial 48-feature set using one-hot encoding. These results are from the initial, pre-cleaning sample, whose target distribution is 74.8% positive / 25.2% negative; the deduplication and record removal in Section 3.2 yield the 4,561-individual analytic sample (74.6% positive) used in all later experiments.

Algorithm	Accuracy	ROC-AUC
Random Forest	0.749	**0.651**
HistGradientBoosting	0.757	**0.651**
Logistic Regression	0.638	0.640
Naive Bayes	0.290	0.612
KNN	0.750	0.599
Decision Tree	0.640	0.569

4.3 Effect of feature encoding

We then ask whether encoding is a meaningful factor in prediction performance. The initial 48-feature baseline was first cleaned as described in Section 3.2, resulting in a 45-feature pre-deduplication core. Removing one additional duplicate construct (Impulsivity = D10Q4) produced the final 44-feature Model A used in the remaining experiments. We then applied the theory-driven hybrid encoding described in Section 3.4 and compared it against a simpler uniform ordinal encoding under identical conditions (the same 44 features, cross-validation folds, and algorithms).

Table 2: Five-fold cross-validation ROC-AUC under the theory-driven hybrid encoding using the 44-feature core set.

Model	CV ROC-AUC
Random Forest	0.650 ± 0.017
Logistic Regression	0.644 ± 0.012
HistGradientBoosting	0.620 ± 0.017

The theory-driven encoding reaches 0.650 with Random Forest (see Table 2), differing from the uniform ordinal encoding by only about 0.015 and suggesting a limited effect of encoding choice.

4.4 Effect of theory-driven feature expansion

We next tested whether theoretically selected feature expansions provided additional predictive value. Models A, B, and C contain 44, 87, and 130 features, respectively, and differ only in their feature sets; preprocessing, encoding, cross-validation folds, and random seed were held constant. Encoding here is fixed to the uniform ordinal scheme, so Model A differs from Table 2 (theory-driven encoding).

As Table 3 reports, adding criminogenic variables in Model B produced only small changes in ROC-AUC, and adding non-criminogenic variables in Model C did not produce a consistent improvement across algorithms. Logistic Regression improved from 0.615 for Model A to 0.621 for Model B but decreased to 0.612 for Model C, while Random Forest performance declined slightly as features were added. HistGradientBoosting showed only a small increase to 0.624. Overall, the theory-driven expansions did not

Table 3: ROC-AUC for theory-driven feature expansion.

Algorithm	A (44)	B (87)	C (130)
Logistic Regression	0.615	0.621	0.612
Random Forest	0.635	0.634	0.631
HistGradientBoosting	0.621	0.622	0.624

overcome the observed performance plateau, suggesting that additional internal PACT features provide limited new predictive information.

4.5 Effect of feature-set size

We also tested whether the plateau was caused simply by having too few intake variables. The feature set was expanded from 45 to 163 variables while excluding institutional-misconduct variables and variables that could introduce data leakage. The baseline in this experiment was the 45-variable pre-deduplication core described in Section 3.3, before the redundant `Impulsivity` variable was removed to form the 44-variable Model A. As reported in Section 3.3, removing this variable did not affect model performance.

Table 4: ROC-AUC after expanding from 45 to 163 features.

Model	ROC-AUC (163)	vs. 45
Random Forest	0.628 ± 0.015	↓
HistGradientBoosting	0.622 ± 0.021	↓
Logistic Regression	0.611 ± 0.008	↓

As shown in Table 4, all three algorithms performed worse with the larger feature set, suggesting that additional intake variables primarily added noise or redundant information rather than useful predictive signal. Together with the theory-driven expansion results, this indicates that adding more information already contained within PACT is unlikely to overcome the observed performance plateau.

4.6 External context and paired robustness

We next tested whether contextual information beyond the core PACT features could provide new predictive signal. Adding the PACT family-poverty variable to Model B produced a small improvement of approximately 0.003 ROC-AUC for Random Forest and HistGradientBoosting, while a broader home-environment feature group provided little additional benefit.

We then added four county-level socioeconomic variables from the American Community Survey (ACS) [12]: poverty rate, median household income, unemployment rate, and high-school completion. In the initial 5-fold evaluation, Random Forest improved from 0.634 to 0.641 and HistGradientBoosting from 0.622 to 0.626, while Logistic Regression remained essentially unchanged. County violent-crime rate from County Health Rankings (CHR) [11] produced almost no additional improvement.

Because these changes were small, we tested their consistency using the paired 10-seed protocol described in Section 3.7. The final external feature set contained 12 county-level socioeconomic and community variables, including food-environment and social-disorganization measures.

Table 5: Paired 10-seed comparison of Model B and B+Geo. B and B+Geo report mean ROC-AUC across seeds, and SD is the standard deviation of the paired improvements.

Algorithm	B	B+Geo	Δ	SD(Δ)	Δ > 0
Logistic Regression	0.615	0.618	+0.0033	0.0012	10/10
Random Forest	0.636	0.642	+0.0066	0.0026	10/10
HistGradientBoosting	0.617	0.631	+0.0133	0.0044	10/10

County-level context improved ROC-AUC for all three algorithms in all 10 seeds (as seen in Table 5), indicating that the improvement was consistent rather than caused by a favorable data split. HistGradientBoosting showed the largest average increase (+0.0133), followed by Random Forest (+0.0066) and Logistic Regression (+0.0033). However, the overall performance remained near the existing ceiling. These results suggest that external contextual information can provide additional predictive value, but the available county-level variables were not sufficient to eliminate the plateau.

4.7 Gang affiliation: a documented null

We also examined whether explicitly adding gang affiliation improved predictive performance. Under the same paired 10-seed protocol, we added `GANGAFF_max`, a gang-affiliation indicator with no missing observations, to Model B.

Table 6: Paired 10-seed comparison of Model B with and without the gang-affiliation indicator.

Algorithm	B	B+Gang	Δ	Δ > 0
Logistic Regression	0.6152	0.6158	+0.0006	9/10
Random Forest	0.6356	0.6354	−0.0002	4/10
HistGradientBoosting	0.6172	0.6163	−0.0009	4/10

As summarized in Table 6, adding the gang flag produced negligible changes in ROC-AUC across all three algorithms. Gang-related information may already be partially represented by two peer items in Model A: `D6AQ2`, which was positive for 1,875 individuals, and `D6BQ3`, which was positive for 597. However, the present analysis does not establish whether this overlap explains the null result. Overall, the standalone gang-affiliation indicator provided little incremental predictive information beyond the existing Model B variables.

4.8 SHAP interpretability

We used SHAP TreeExplainer [6, 7] to interpret the best-performing model, Random Forest with Model B+Geo (99 features). We report feature importance as the mean absolute SHAP value over a sample of 1,500 individuals. The highest-ranked individual features included `D1Q6` (against-person felony history), `SUBABUSE_max` (substance-use severity), `D2Q1` (gender), `TRUANCY_max`, and `D1Q8`

Table 7: Domain-level SHAP importance for Random Forest with Model B+Geo.

Domain	$\sum$ |SHAP|	Domain	$\sum$ |SHAP|
Attitudes / skills	0.052	Criminal history	0.049
County context	0.030	School	0.025
Peers	0.023	Substance	0.017
Family	0.017	Gender	0.010
Intake risk score	0.001		

Table 8: Factors explaining why theoretically relevant variables did not improve prediction performance.

Potential explanation	Examples
Restriction of range	Family, school, and cognitive-skill variables in an already-incarcerated population
Redundancy / overlap	Duplicate/correlated features; county crime rate after socioeconomic variables
Limited intake information	All prediction models
Near-constant / low variance	`ADJCINS_max`, `RISKSEX_max`, etc.
High missingness	Mental-health variables in `D9B*`
Non-criminogenic factors	Mental health, trauma, and self-esteem

(sex-offense history). The highest-ranked external feature was county median income at rank 12.

Attitudes/skills and criminal history had the highest domain-level SHAP importance, Table 7, followed by county context, indicating that external county variables contributed to predictions. The precomputed intake risk `Score` added little once individual assessment items were included. Because correlated predictors can redistribute SHAP importance and domain sums depend partly on domain size, these rankings should be interpreted cautiously.

5 Discussion

Our experiments identified several reasons why theoretically relevant variables may contribute little to prediction performance. As summarized in Table 8, some variables contain information already captured by other features, while others have low variance, high missingness, or limited variation because the dataset contains only juveniles already involved in the justice system. Non-criminogenic factors may also be important for treatment without strongly improving general recidivism prediction.

Across the classical models and encodings evaluated, internal PACT modifications produced little improvement, whereas county context yielded small, consistent gains. This suggests that the plateau reflects limited intake information more than model choice, although we did not test deep neural networks or learned representations. Given their moderate performance, these models should support, not determine, individual decisions. We did not assess fairness across racial, gender, or socioeconomic groups. Individual-level predictors were limited to intake-time information; county variables represented external administrative and community context.

6 Conclusion

Our results suggest that the performance plateau in PACT-based juvenile recidivism prediction is more closely associated with limitations in available intake information than with algorithm, encoding, or feature-set choices. Expanding PACT features produced little improvement, while county-level contextual variables yielded small but consistent ROC-AUC gains. SHAP analysis highlighted attitudes and skills, criminal history, and county context as influential domains, whereas the aggregate PACT risk score contributed little beyond its underlying items. These findings motivate incorporating richer, temporally appropriate contextual information while emphasizing that, given moderate predictive performance and the consequential nature of juvenile justice decisions, such models should support rather than replace professional judgment. Future work should examine richer contextual data, calibration, and fairness.

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